

**Electrodynamics(H2) (SC1.102a)**  
**IIIT-H, Semester Winter 26, Assignment 2**

*Submission deadline: 8<sup>th</sup> May 2026 (strict deadline)*

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1. There are many worked out examples and exercises in the Griffiths book. Try to do as many as possible for practice.
2. There is a grounded conducting sphere of radius  $R$ . A point charge  $q$  is placed outside the sphere at a distance  $a$  from the center. Find the potential at any point outside the sphere. Also find the attraction between the charge and the sphere. *Method of image. The question was partially done in the class. Marks will be deducted for not showing all steps and for lack of explanations.*
3. Consider four charges:  $+q$  at  $(a, a)$ ,  $-q$  at  $(-a, a)$ ,  $+q$  at  $(-a, -a)$  and  $-q$  at  $(a, -a)$ . Calculate the potential at  $r \gg a$ . Show that the monopole and the dipole term vanishes, and the leading term of the potential scales as  $1/r^3$ .
4. Show that when a electric dipole  $\mathbf{p}$  is placed in a uniform electric field  $\mathbf{E}$  it experiences a torque  $\mathbf{N} = \mathbf{p} \times \mathbf{E}$ . Show that the energy stored in a dipole is  $-\mathbf{p} \cdot \mathbf{E}$ . Show that if the field is not uniform then there is a net force on the dipole. Calculate the expression of the force.
5. When a dielectric material is placed in an external electric field  $\mathbf{E}$ , an induced dipole moment  $\mathbf{p} = \alpha \mathbf{E}$  develops, where  $\alpha$  is the atomic polarizability. For the hydrogen atom in its ground state, the electron charge density is given by

$$\rho(r) = \frac{-e}{\pi a^3} e^{-2r/a}, \quad (1)$$

where  $e$  is the magnitude of the electron charge and  $a$  is the Bohr radius. According to the classical model of Hydrogen atom the electron cloud remains a fixed, spherically symmetric charge distribution. Under an external electric field  $\mathbf{E}$ , the nucleus is displaced by a small distance  $d$  relative to the center of the electron cloud. The system reaches equilibrium when the external force on the nucleus is balanced by the electrostatic restoring force due to the electron cloud. Answer the following questions

- (a) Calculate the electric field  $E_e(r)$  produced by the electron cloud at a distance  $r$  from the center.
  - (b) For small displacement  $d$ , determine the restoring field acting on the nucleus.
  - (c) Using the equilibrium condition, find  $d$  in terms of  $E$ .
  - (d) Hence compute the induced dipole moment  $p = ed$ , and obtain the polarizability  $\alpha$ .
6. Find the magnetic field inside and outside of a long solenoid, consisting of  $n$  closely wound turns per unit cylinder length. Radius of the cylinder is  $R$  and the steady current through the wire is  $I$ .
  7. If  $\mathbf{K}$  is the surface current then the expression of magnetic dipole moment is

$$\mathbf{m} = \frac{1}{2} \int \mathbf{r} \times \mathbf{K}(\mathbf{r}) da.$$

Using this formula, calculate the magnetic dipole of a rotating spherical shell of radius  $R$  and uniform charge density. Calculate the potential at a distance  $\gg R$ . *Hint:  $\int r_i r_j da = \frac{4\pi R^4}{3} \delta_{ij}$ .*

8. EM waves inside a conductor.

- (a) Obtain the electromagnetic wave equations for  $\vec{E}$  and  $\vec{B}$  in a conducting media with free charge density  $\rho$  and current density  $\vec{J}$ . Take the conductivity as  $\sigma$ .
- (b) Assuming plane wave solutions, show that both  $\vec{E}$  and  $\vec{B}$  are attenuated inside the conductor.